moepager

Expert-Aware Page-Cache Management for Mixture-of-Experts LLM Inference on Memory-Constrained Linux Machines

Formal title: System and Method for Non-Invasive, Expert-Aware Page-Cache Management for Mixture-of-Experts Model Inference on Memory-Constrained Linux Hosts

Repository: https://github.com/shreshth006/moepager · team fork: https://github.com/AyushSinha2210/moepager (branch `ayush`)

1. Title, Abstract and Architecture

1.1 Abstract

Mixture-of-Experts (MoE) language models activate only a few "expert" sub-networks per layer for each token: 8 of 64 in OLMoE-1B-7B, 4 of 32 in gpt-oss-20b, 8 of 128 in Qwen3-30B-A3B. That makes them attractive for consumer machines, which run them with llama.cpp in `mmap` mode when the model is larger than free RAM, leaving the Linux page cache to decide which weights stay in memory.

We show that the kernel's generic behaviour fits this workload poorly: - **Scattered units.** One expert's weights are three slices in three tensors hundreds of megabytes apart. - **Read-around pollution.** On a page fault, the kernel reads a window around the faulting page: 4 MiB on our test machine's btrfs. That pulls unused neighbouring experts into the cache and multiplies SSD traffic 3.7–4.8×. - **Repacking.** llama.cpp's default CPU weight repacking moves 77–100 % of expert bytes into anonymous memory, outside the page cache.

We present **moepager**, a Rust toolchain and an unprivileged user-space daemon that manage the shared page cache at the granularity of (layer, expert) units, without changing the inference engine's code or binary: 1. It parses the GGUF header into an exact expert map. 2. It observes expert misses black-box by polling sentinel pages with `mincore()`. 3. It answers each miss with **expert-completion readahead**: bulk `posix_fadvise(WILLNEED)` of the expert's remaining slices in 128 KiB chunks. The chunking is necessary because the kernel silently truncates larger requests. 4. A 64-line `LD_PRELOAD` shim turns off fault read-around on the model mapping, which no stock llama.cpp option does on single-node machines.

The evaluation toolchain records ground-truth routing from llama.cpp, computes exact LRU miss-ratio curves, simulates policies against a Belady optimum, and runs llama.cpp under cgroup memory limits.

On a 15 GB laptop running OLMoE-1B-7B (4.2 GB) under memory limits of 1.5 GB and 2.5 GB, moepager's configuration decoded **1.40 and 3.21 tokens/s**, versus 0.35 and 0.87 tokens/s for the same engine served by the default page cache. That is **3.7–4.0× faster**, with **7–8× less SSD traffic** and memory-pressure stall reduced from 12.7 % to 1.0 %. Default llama.cpp, by contrast, was OOM-killed at both limits unless allowed to spill 2.3–2.5 GB into compressed RAM swap. On a second model, gpt-oss-20b (12.1 GB), replaying real routing under 4 GB and 6 GB limits showed the same pattern: 1.6–2.0× faster than the kernel default, with 15–20× less SSD traffic.

We also report what did not work: a value-based pinning policy captured little of the optimal policy's 34–74 % miss reduction, and completion readahead without turning off read-around made performance worse.

1.2 System architecture

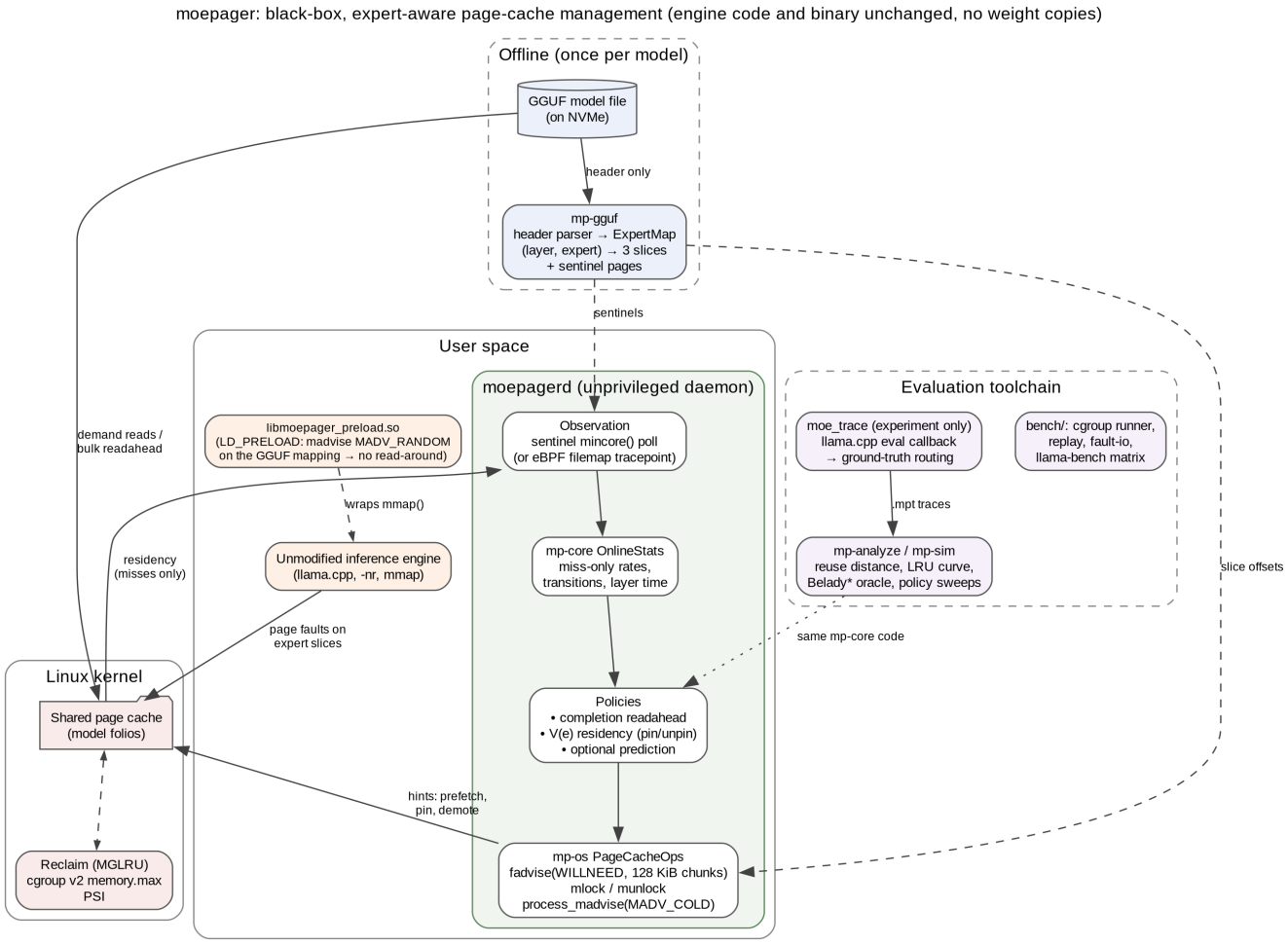


Figure 1. moepager architecture (source: `docs/report/architecture-portrait.dot`; a landscape version is `architecture.png`).

- **Offline, once per model:** `mp-gguf` reads only the GGUF header and builds the expert map: each (layer, expert) unit's three slice offsets and sentinel pages.
- **Runtime:**
  - The unmodified engine runs with `--no-repack` (so weights stay in the page cache) and with the `LD_PRELOAD` shim, which turns off read-around on the model mapping.
  - Its page faults bring expert slices into the shared page cache.
  - `moepagerd` polls sentinel pages, so it sees misses only. It updates `mp-core` statistics and issues page-cache hints through `mp-os`:
    - chunked WILLNEED for completion readahead (no privilege needed);
    - optional `mlock` pinning (`CAP_IPC_LOCK`);
    - optional `MADV_COLD` demotion (`CAP_SYS_NICE`).
  - No weights are copied: engine and daemon share the same page-cache pages.
  - Both run inside one cgroup v2 scope with a `memory.max` budget, so every page is charged to the same budget.
- **Evaluation:**
  - `moe_trace`, an experiment-only llama.cpp eval callback, records ground-truth routing.
  - `mp-analyze` and `mp-sim` evaluate policies against Belady*, using the same `mp-core` code as the daemon.

- `bench/` runs real-kernel replays, the bandwidth microbenchmark and llama-bench under memory limits.

2. Introduction and Problem Statement

2.1 Background

Mixture-of-Experts (MoE) language models replace each dense feed-forward block with many smaller "expert" networks plus a router. For every token, the router picks a few experts per layer (top-k), so only a small fraction of the parameters is used per token. In the three models studied here (values read from the real model file headers):

Model	File (quantisation)	Size	Layers × experts	top-k	Expert share of file
OLMoE-1B-7B-0924	Q4_K_M	4.21 GB	16 × 64	8	92.6 %
gpt-oss-20b	MXFP4	12.11 GB	24 × 32	4	84.1 %
Qwen3-30B-A3B	Q4_K_M	18.56 GB	48 × 128	8	94.6 %

Because only 4–8 experts per layer are touched per token, MoE models are an attractive fit for consumer machines. A common way to run a model larger than free RAM is llama.cpp in `mmap` mode: the model file is mapped into memory, and the Linux page cache loads weights on demand and evicts them under memory pressure.

2.2 Problem statement

The kernel's page cache is generic. Its reclaim policy (MGLRU) and its fault read-around know nothing about how a GGUF file is laid out or how MoE routing behaves. Our measurements on a 15 GB laptop (Section 6) show three concrete problems:

1. **The unit of use is scattered.** One expert at one layer is stored as three slices in three different tensors (`ffn_up_exps`, `ffn_gate_exps`, `ffn_down_exps`), hundreds of megabytes apart. The kernel cannot know these three ranges belong together.
2. **Read-around pollutes the cache.** On a page fault, Linux reads a window around the faulting page: 4 MiB on this machine's btrfs filesystem. For expert slices of about 1–1.3 MB, most of that window belongs to neighbouring experts that will not be used. Replaying real OLMoE routing under a 1.5 GB budget, the SSD read **1.80 GB per token to deliver 0.49 GB of needed expert weights** (3.7× amplification), and almost every expert access missed.
3. **Fault-driven reads are slow.** Faulting cold experts in through mmap delivered only 0.15–0.24 GB/s on the real model file. Reading the same bytes with large prefetch requests delivered 1.5–1.7 GB/s.

A further, practical problem: llama.cpp on x86 CPUs with AVX2 by default *repacks* the most common quantised expert weights into ordinary (anonymous) memory. That is 77 % of the expert bytes for OLMoE and Qwen3 Q4_K_M, and 100 % for gpt-oss MXFP4. Under memory pressure those pages go to swap instead of the page cache.

Goal: make MoE inference faster on memory-constrained Linux machines, using knowledge of the model file's structure and of routing behaviour, **without modifying the inference engine** and without copying the weights.

3. Literature Review and Objectives

3.1 Related work

MoE offloading inside the inference engine. Most systems that keep only part of an MoE model in fast memory are built into the engine, where the router's decisions are directly visible: - mixtral-offloading (Eliseev & Mazur, 2023): LRU expert cache plus speculative prefetch. - MoE-Infinity (Xue et al., 2024): activation-aware expert cache. - HOBBIT (2024): mixed-precision experts, implemented as about 8,000 lines added to llama.cpp. - Pre-gated MoE, ProMoE, DuoServe-MoE, Fate and similar: predict next-layer experts and prefetch them.

Inside llama.cpp itself there are several open proposals: PR #25294 (O_DIRECT expert streaming that bypasses the page cache), PR #26824 (expert heat map and mmap pinning), PR #29887, issue #20757 and discussion #27149. All of these modify the engine.

Using the OS page cache as the expert tier. Si et al. (arXiv 2608.12103, 2026) is the closest study. On a datacenter GH200 node they show that kernel recency (LRU) serves about as much demand as an oracle static-frequency pin, that Belady's bound leaves about one third of LRU's misses addressable, and that `fadvise(WILLNEED)` prefetch hints gained only 0.3–5 %. Their engine reads with `pread`, so mmap fault behaviour and file structure play no role.

Operating-system mechanisms. - cache_ext / cachebpf (Zussman et al., 2025) lets eBPF programs replace page-cache eviction, but needs a patched kernel. - MGLRU and DAMON are generic and know nothing of model structure. - vmtouch pins or evicts file ranges given by the user. - "LLM in a flash" (Alizadeh et al., Apple, 2024) and PowerInfer-2 bundle co-used weights to make flash reads large. That is the same idea as our expert-completion readahead, but those systems implement it inside the engine and require reformatting the model.

Gap. We found no system that manages the page cache for an *unmodified* engine using the *model file's structure*, and none that measures the effect of mmap fault read-around on MoE experts.

3.2 Objectives

- **O1.** Build an exact map from GGUF files to (layer, expert) units and their byte and page ranges, verified on real models.
- **O2.** Observe expert usage from outside the engine, without root.
- **O3.** Build an evaluation toolchain: ground-truth routing traces, trace analysis, a policy simulator with an optimal (Belady) bound, real-kernel replay, and benchmarks under a fixed memory budget.
- **O4.** Implement page-cache actuators: chunked readahead, pinning and demotion. Combine them in a daemon whose policy code is identical to the simulated one.
- **O5.** Decide go/no-go on real hardware with predefined kill criteria, and report only measured numbers.

4. Methodology and Algorithms

4.1 Expert map (from the GGUF header)

A GGUF file starts with a header listing every tensor's name, shape, type and offset. `mp-gguf` parses only this header, so a truncated prefix of a file works too. For each tensor named `blk.<L>.<ffn_up|ffn_gate|ffn_down|ffn_gate_up>_exps.weight` with shape `[ne0, ne1, n_expert]`:

- `bytes = type_size × ne0·ne1·n_expert / block_size`, using the ggml type table.
- Expert `e`'s slice is `[offset + e·bytes/n_expert, offset + (e+1)·bytes/n_expert)`. Each expert is contiguous inside a tensor because the expert index is the outermost dimension.
- 2-D per-expert tensors (`*_exps.bias`, `[ne0, n_expert]`) contribute one row per expert.

The slices of the same (L, e) are grouped into a **unit**, the granule for everything that follows. Slices are 32-byte aligned rather than page aligned, so a boundary page can belong to two units. A reverse page→unit index (sorted intervals plus a prefix-maximum array) handles this in O(log n).

For each slice, one **sentinel page** is chosen in its middle, at least 128 KiB from either edge, so that a neighbour's 128 KiB read-around cannot trigger it.

4.2 Black-box observation

The daemon maps the model file read-only and periodically calls `mincore()` on the sentinel pages. This works without root, even for files it doesn't own. A sentinel changing from non-resident to resident means "unit (L, e) has just been faulted in", which is a **miss**.

Hits are invisible to user space: pages already mapped are read through the CPU's page tables with no kernel involvement. All online statistics are therefore computed from misses only (Section 4.4).

Token boundaries are not visible either. They are inferred from layer order: a new token starts when the observed layer drops below the running maximum minus a slack. With root, the same events can be taken from the `filemap:mm_filemap_add_to_page_cache` tracepoint (bpftrace script included).

4.3 Expert-completion readahead (the main mechanism)

When a miss of unit u is observed, the daemon immediately issues `posix_fadvise(WILLNEED)` over **all** slices of u, so the parts the engine has not reached yet (typically the gate and down slices, hundreds of MB away) arrive as large asynchronous reads instead of many small synchronous faults. Two details turned out to be essential:

1. **Chunking.** The kernel silently truncates a single WILLNEED to `max(bdi->io_pages, ra->ra_pages)` pages (`force_page_cache_ra()` in `mm/readahead.c`). We found this when CI populated exactly 256 KiB of a 1 MiB request. Requests are therefore issued in 128 KiB chunks.
2. **Refractory window.** Each unit has three sentinels, and the daemon's own prefetch flips the other two. Repeated misses of a unit within 20 ms are ignored so that each real miss causes one prefetch.

The mechanism is prediction-free: it acts only on experts the router has already chosen.

4.4 Residency policy V(e) (optional, privilege-dependent)

For pinning with `mlock`, each unit e of size s_e gets a value per byte:

```
    V(e) = rate(e) · (t_fault + s_e / B_demand) / s_e
```

Units are pinned greedily in decreasing V order until the pin budget is used up. Hysteresis (a newcomer must beat a pinned unit by 25 %) limits churn.

Because only misses are observable, rate(e) is defined as the **miss rate per token while unpinned**: decayed miss count ÷ decayed number of tokens the unit was exposed (not pinned). That is exactly the counterfactual a pin decision needs. While a unit is pinned it produces no misses and no exposure, so its estimate is frozen. A small random fraction of pinned units is released at each re-plan ("exploration") to

refresh stale estimates. Exponential decay uses the scaled-weight trick, with periodic renormalisation, for O(1) updates.

Demotion of cold units uses `process_madvise(pidfd, MADV_COLD)` on the engine's mapping, because `fadvise(DONTNEED)` cannot evict pages another process has mapped (verified experimentally).

4.5 Simulation and the optimal bound

`mp-sim` replays a routing trace against a byte-capacity cache: - A hit costs nothing. - A miss is queued on a FIFO I/O channel, costing `t_fault + s/B_demand`, or `s/B_bulk` when completion readahead is on. - Compute time per layer is taken from the trace.

Policies: LRU, LFU, prefix pinning, an oracle static-frequency pin, V(e) with full or miss-only observation, predictive prefetch, and **Belady*** (evict the unit whose next use is furthest in the future). Belady* is the optimal bound for equal sizes and the standard heuristic for variable sizes.

Independently, `mp-analyze` computes reuse distances with a Fenwick tree in O(N log N) and derives the **exact LRU miss-ratio curve**. An access hits in a byte-capacity LRU cache iff (bytes of distinct units since its last use) + its own size ≤ capacity. The simulator's LRU agrees with this curve exactly, a cross-check between two independent implementations.

4.6 Experimental method (go/no-go)

Kill criteria were fixed before measuring (IDEA_REVIEW §6): - **K1:** Belady* cannot save at least 15 % of misses relative to the kernel. - **K2:** fault-driven bandwidth is at least 0.75× bulk bandwidth. - **K3:** default llama.cpp beats every page-cache configuration.

Experiments:

- **A: routing.** Record real routing with `moe_trace`, a 170-line experiment-only tool that reads llama.cpp's `ffn_moe_topk-<layer>` tensors through the scheduler's eval callback. Three prompts (chat, code, prose) × 255 decode tokens on OLMoE, then analyze and simulate.
- **B: real-kernel replay.** Replay the prose trace on the real model file under a cgroup v2 memory wall (`systemd-run --user -p MemoryMax=…`, swap 0). Measure SSD bytes (`/proc/self/io`), major faults and time.
- **C: bandwidth microbenchmark.** Read cold expert-sized units of the real file through mmap faults, chunked WILLNEED, or pread.
- **D: end to end.** `llama-bench` decode throughput under the same memory walls, for:
  - B0: default llama.cpp;
  - B2: `--repack 0`;
  - D2: B2 + moepagerd;
  - P: B2 + read-around off;
  - PD: P + moepagerd.

  Daemon and engine always share the cgroup, so page-cache charges count against the same budget.

5. Implementation

5.1 Technology stack and structure

Layer	Choice	Why
Core, simulator, daemon, CLI	**Rust** (stable), 10 crates, ≈ 8,900 lines	Predictable latency for a daemon; fast enough for simulation sweeps; one language, so the daemon and the simulator share the *same* policy code
OS interface	`libc` syscalls behind traits (`PageCacheOps`, `ResidencyProbe`)	Real Linux implementation + `MockOps` for deterministic tests
Ground-truth capture	C++ against `libllama` (`tools/moe_trace`, 169 lines)	Needs the engine's eval callback; experiment-only
Read-around shim	C, `LD_PRELOAD` (`tools/preload`, 64 lines)	Disables fault read-around on the model mapping without touching the engine's code or binary
Privileged tracing	bpftrace script (26 lines)	Optional, root only
Fixtures, metric parsing, probes	Python (stdlib + pytest), 674 lines	Independent GGUF writer so parser tests aren't self-confirming
Experiments	Bash + cgroup v2 (`systemd-run --user`), no root	Fixed memory budget per run
CI	GitHub Actions: fmt, clippy `-D warnings`, tests, ruff, pytest, demo smoke run	

Crate	Responsibility
`mp-gguf`	GGUF v2/v3 header parser, ggml type table, `ExpertMap`, page index, sentinels, repack-risk detection
`mp-trace`	`.mpt` binary trace format (spec in `docs/TRACE_FORMAT.md`), CSV import/export, token inference
`mp-core`	Deterministic RNG, cost model, `OnlineStats` (miss-only), `ResidencyEngine` (V(e)), `PrefetchPlanner`
`mp-synth`	Synthetic routing generator (skew, reuse, cross-layer affinity, drift)
`mp-analyze`	Fenwick-tree reuse distance, exact LRU miss-ratio curve, routing predictability, timing
`mp-sim`	Trace-driven cache + I/O-channel simulator, policies, Belady*
`mp-os`	`mincore`, `cachestat`, chunked `fadvise`, `mlock`, `process_madvise`, `/proc/pid/maps`/`smaps` parsing
`mp-recorder`	Sentinel and full-scan recorders, page→expert conversion, bpftrace ingest, mmap/pread trace replayer
`moepager`	CLI: `gguf-map`, `synth`, `analyze`, `sim`, `import-csv`, `trace-csv`, `record`, `ingest-bpftrace`, `page2expert`, `replay`, `fault-io`
`moepagerd`	Daemon: policies `none`, `willneed-all`, `lru-pin`, `v`; sources `sentinel` / `trace:`; ops `linux` / `mock`

5.2 Functional completeness

Feature	Status
GGUF expert map, verified on 3 real models	done, tested
Ground-truth routing capture + import	done, used on OLMoE
Trace analysis, simulator, Belady bound	done, tested, used
Unprivileged sentinel observation	done, tested end to end on real files
Chunked completion readahead	done, measured (Section 6)
Read-around shim	done, measured (Section 6)
V(e) pinning	implemented and unit-tested; live use needs `CAP_IPC_LOCK` (default memlock limit is 8 MB), **not measured**
`MADV_COLD` demotion	implemented; needs `CAP_SYS_NICE`, **not measured**
eBPF observation	script + ingest written; needs root, **not run**
Start-up checks: read-around still on, daemon outside the engine's cgroup	done, tested live
`MADV_POPULATE_READ` prefetch fallback (`--prefetch-mode populate`)	done, tested on a real file; not benchmarked
PSI-driven budget control, dashboard	future work

5.3 Correctness

- **Tests:** 87 Rust and 13 Python tests, run in CI. They include:
  - parser checks against GGUF files written by an *independent* Python writer;
  - an assertion that simulated LRU equals the analytically computed LRU curve exactly;
  - a check that Belady* never loses to any other policy;
  - real-kernel tests of `mincore`/`cachestat`/`fadvise`/`mlock` semantics on a real file;
  - end-to-end recorder tests that recover a replayed trace with 100 % recall;
  - daemon dry runs.
- **Real-model checks:**
  - The map of OLMoE/Qwen3/gpt-oss matches tensor sizes in the headers.
  - Each captured (token, layer) has exactly top-k experts.
  - The no-read-around replay reproduced the simulator's LRU prediction (Section 6.3).
- **Bugs found by testing and fixed:**
  - WILLNEED truncation (found by CI);
  - the btrfs device-number mismatch between `stat()` and `/proc/pid/maps`;
  - read-around/compression spill causing false expert detections;
  - duplicate completion prefetches (refractory window).

5.4 Demonstration

`make demo` runs the full pipeline with no model or root. It parses the real Qwen3 map, generates routing, analyzes it, simulates policies, does a daemon dry run, and records a replayed trace black-box. With a model, `bench/phase7.sh` runs experiments A–D. Terminal screenshots of the tools on the real OLMoE model follow.

5.5 Screenshots (real output on OLMoE-1B-7B)

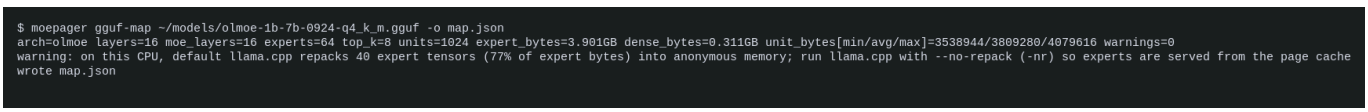


Figure 2. Expert map from the real GGUF header, with the repack warning for this AVX2 CPU.

```
$ ./build/moe_trace/moe_trace -m olmoe-1b-7b-0924-q4_k_m.gguf -o trace-prose.csv -n 256 --no-repack -p "..."
moe_trace: 256 tokens generated, 32640 expert rows -> bench/results/phase7-olmoe/trace-prose.csv
token,layer,expert,t_ns
0,0,6,1106548300
0,0,15,1106548300
0,0,63,1106548300
...
```

Figure 3. Ground-truth routing capture through llama.cpp's eval callback.

```
$ moepager import-csv trace-prose.csv --map map.json -o trace-prose.mpt
imported 32640 rows, 255 tokens -> trace-prose.mpt

$ moepager analyze trace-prose.mpt --map map.json
source=llama-evalcb observation=Full tokens=255 accesses=32640
footprint=2.866GB cold_misses=750 median_reuse_units=Some(193)
token_reuse=0.448 top10_share=0.376 norm_entropy=0.832 units_touched=0.732
next-layer recall@k: transitions=0.687 popularity=0.453 prev-token=0.455 (concentration top4=0.399)
layer gap p10/p50/p90 = 1.507/1.936/3.224 ms; token p50 = 35.624 ms
LRU miss-ratio curve (capacity as % of footprint):
    10%  cap=0.287GB  miss_ratio=1.000  miss_bytes/token=0.488GB
    20%  cap=0.573GB  miss_ratio=0.554  miss_bytes/token=0.271GB
    30%  cap=0.860GB  miss_ratio=0.454  miss_bytes/token=0.223GB
    40%  cap=1.146GB  miss_ratio=0.321  miss_bytes/token=0.157GB
    50%  cap=1.433GB  miss_ratio=0.286  miss_bytes/token=0.140GB
    60%  cap=1.719GB  miss_ratio=0.260  miss_bytes/token=0.127GB
    70%  cap=2.006GB  miss_ratio=0.212  miss_bytes/token=0.104GB
    80%  cap=2.292GB  miss_ratio=0.139  miss_bytes/token=0.068GB
```

Figure 4. Importing the routing trace and analyzing it: reuse, predictability, exact LRU curve.

```
$ moepager sim trace-prose.mpt --map map.json --caps 0.25,0.5 --completion off --policies lru,lfu,static-freq,v-full,v-miss,belady
|trace=bench/results/phase7-olmoe/trace-prose.mpt tokens=255 (warmup 16) expert_bytes=3.901GB t_layer=1.936ms demand=0.45GB/s bulk=2.4GB/s
| capacity | completion | policy | miss ratio | SSD GB/token | est tok/s
|---|---|---|---|---|---
| 25% (0.98 GB) | off | lru | 0.434 | 0.212 | 1.96
| 25% (0.98 GB) | off | lfu | 0.331 | 0.162 | 2.51
| 25% (0.98 GB) | off | static-freq* | 0.375 | 0.185 | 2.23
| 25% (0.98 GB) | off | v-full | 0.375 | 0.184 | 2.24
| 25% (0.98 GB) | off | v-miss | 0.391 | 0.193 | 2.15
| 25% (0.98 GB) | off | belady* | 0.237 | 0.116 | 3.40
| 50% (1.95 GB) | off | lru | 0.222 | 0.108 | 3.61
| 50% (1.95 GB) | off | lfu | 0.080 | 0.039 | 8.27
| 50% (1.95 GB) | off | static-freq* | 0.149 | 0.073 | 5.06
| 50% (1.95 GB) | off | v-full | 0.146 | 0.071 | 5.16
| 50% (1.95 GB) | off | v-miss | 0.151 | 0.074 | 5.01
| 50% (1.95 GB) | off | belady* | 0.058 | 0.028 | 10.32
```

Figure 5. Policy simulation, including the Belady bound.*

```
$ python3 python/moepager_tools/collect.py bench/results/phase7-olmoe   (final pass, 3 reps)
| budget | config | tok/s | SSD GB/token | majflt/token | refault/token | PSI some avg10 | PSI stall ms | peak GB |
|---|---|---|---|---|---|---|---|---|
| D-1500M | B0 | 3.17 | 0.310 | 38205.5 | 64181.4 | 16.21 | 6351 | 1.57 |
| D-1500M | B0S | TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| D-1500M | B2 | 0.35 | 2.871 | 2112.4 | 687991.5 | 12.74 | 34763 | 1.57 |
| D-1500M | D2 | 0.44 | 2.899 | 870.6 | 695946.5 | 4.76 | 11368 | 1.57 |
| D-1500M | P | 0.74 | 0.250 | 50418.0 | 50474.9 | 18.14 | 24856 | 1.57 |
| D-1500M | PD | 1.40 | 0.337 | 23901.5 | 71515.3 | 1.00 | 1626 | 1.57 |
| D-2500M | B0 | 7.89 | 0.129 | 13853.5 | 20049.7 | 12.03 | 3205 | 2.62 |
| D-2500M | B0S | TBD | TBD | TBD | TBD | TBD | TBD | TBD |
| D-2500M | B2 | 0.87 | 1.189 | 1402.1 | 278880.5 | 14.82 | 17428 | 2.62 |
| D-2500M | D2 | 1.25 | 1.014 | 515.1 | 236641.7 | 12.26 | 9860 | 2.62 |
| D-2500M | P | 2.51 | 0.104 | 14496.8 | 14548.9 | 4.45 | 3756 | 2.62 |
| D-2500M | PD | 3.21 | 0.166 | 8160.5 | 29773.9 | 2.51 | 2116 | 2.62 |

# moepagerd counters, PD @ 2500M:
{ "events": 11938, "tokens": 98, "prefetch_ops": 11938, "pin_ops": 0, "unpin_ops": 0, "demote_ops": 0, "failed_ops": 0, "budget_clamps": 0, "suppressed_prefetches": 0}
moepager_preload: MADV_RANDOM on 4018 MiB mapping (fd 3): ok
```

Figure 6. Collected end-to-end results (B0S rows are OOM-killed runs), daemon counters, and the shim at work.

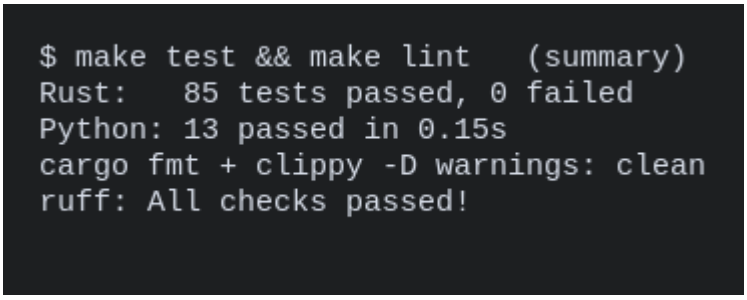


Figure 7. Test and lint status.

6. Results and Performance Evaluation

6.1 Setup

Item	Value
Machine	Laptop, Intel i5-12450H (8 cores / 12 threads, AVX2), 15 GiB RAM + 8 GiB zram swap, desktop session running
Storage	Solidigm P41 Plus 512 GB NVMe (PCIe 4, QLC); `/home` is btrfs with `compress=zstd:1`; btrfs readahead window 4 MiB
OS	Fedora 42, Linux 6.19.14, MGLRU enabled
Engine	llama.cpp commit 970b51c (2026-10-22), CPU only, 8 threads
Model	OLMoE-1B-7B-0924 Q4_K_M (4.21 GB; 16 layers × 64 experts, top-8; experts 3.90 GB, units ≈ 3.8 MB)
Memory budget	cgroup v2 `memory.max` = 1.5 GB or 2.5 GB, swap off unless stated; daemon and engine in the same cgroup
Cold start	model evicted from the page cache before every run

All numbers below are measured. Raw outputs are in `bench/results/phase7-olmoe/` (logs, CSVs, cgroup statistics, journal excerpts).

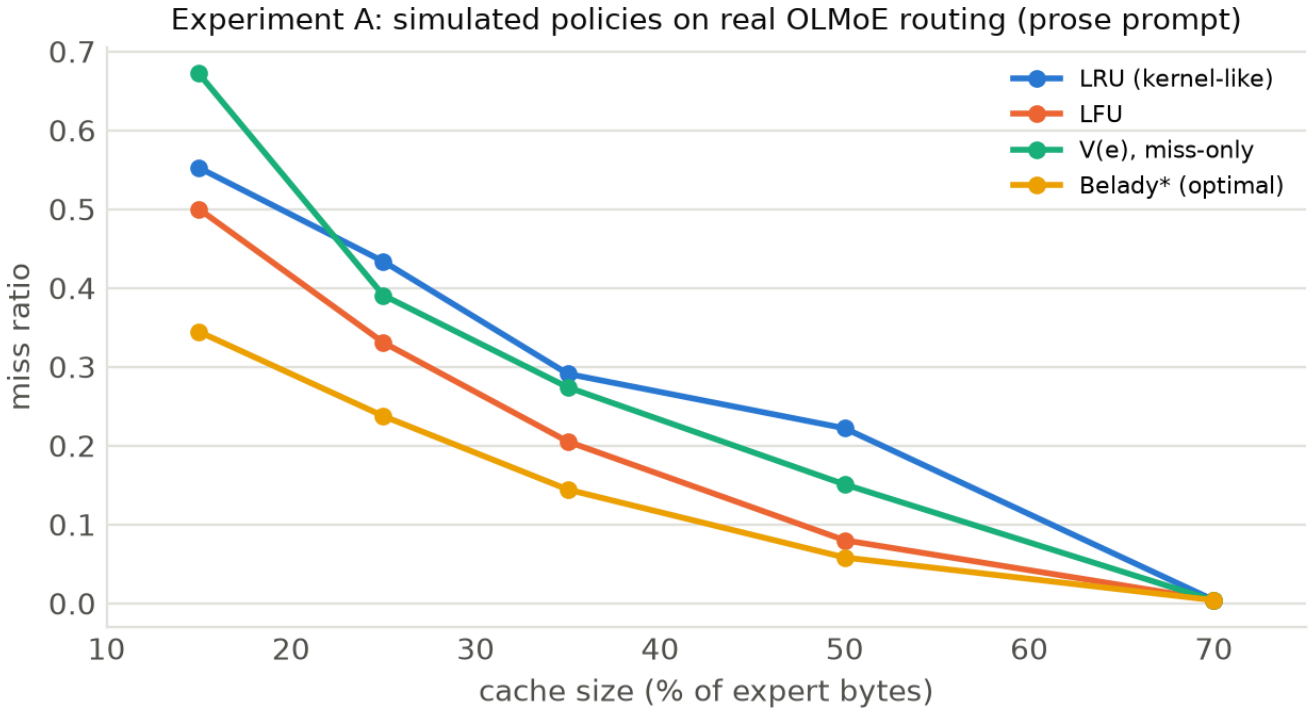
6.2 Experiment A: real routing, and how much any cache policy could gain

`moe_trace` recorded 255 decode tokens for each of three prompts: 32,640 expert uses per prompt. Measured properties:

Prompt	Same expert reused on the next token	Top-10 % experts' share of uses	Next-layer recall: transitions / popularity / previous token
chat	41 %	30 %	0.60 / 0.36 / 0.42
code	48 %	55 %	0.78 / 0.66 / 0.48
prose	45 %	38 %	0.69 / 0.45 / 0.46

Routing is strongly local in time (41–48 % reuse) and predictable across layers (60–78 % recall). The simulator was fed these traces with expert caches of 15–70 % of the expert bytes. Miss ratios for the prose prompt (other prompts in the CSVs):

Cache size	LRU	LFU	V(e), miss-only	Belady* (optimal)	Belady* saves vs LRU
15 %	0.552	0.500	0.672	0.345	38 %
25 %	0.434	0.331	0.391	0.237	45 %
35 %	0.291	0.205	0.274	0.144	50 %
50 %	0.222	0.080	0.151	0.058	74 %
70 %	0.004	0.004	0.004	0.004	4 %



Reading: - An optimal policy would avoid 34–74 % of LRU's misses at 15–50 % capacity across all three prompts, so kill criterion **K1 does not fire**. - Our V(e) residency policy captures only part of that gap. It beats LRU at 25–50 % but is *worse* at 15 %. - Plain LFU beats V(e) in most cells.

Why V(e) lagged (pin-fraction sweep). The default lets V(e) pin only 60 % of the cache and leaves the rest to LRU. With full observation and a pin fraction of 0.95, V(e) matches LFU almost exactly: prose at 50 %, 0.081 vs 0.080; code at 15 %, 0.278 vs 0.276. Its ranking is sound.

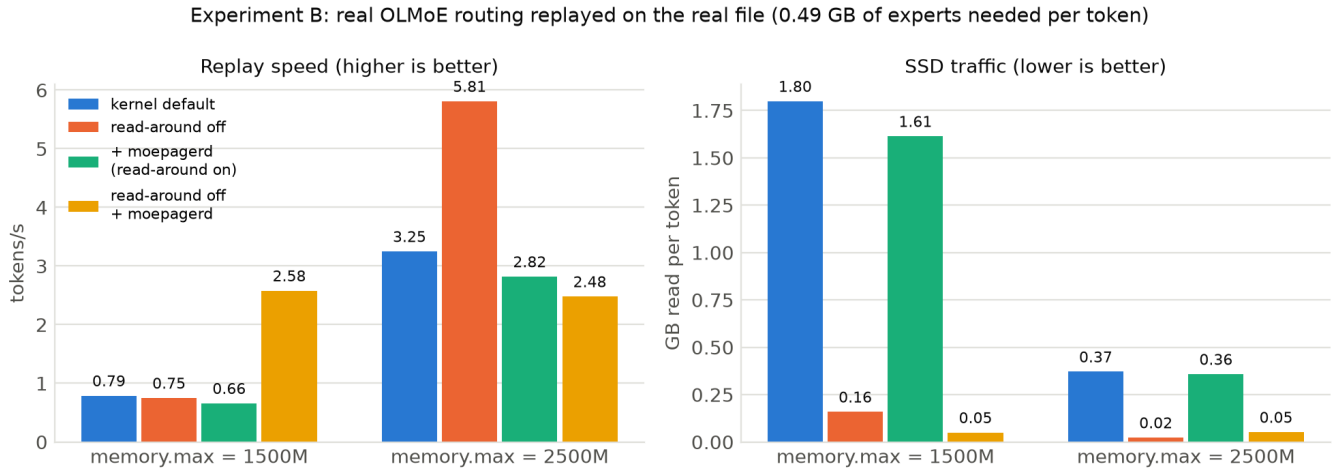
The remaining gap is the **cost of black-box observation**. With misses only, V(e) at 0.95 has 10–40 % more misses than with full observation in most cells, and up to 2× in a few code-prompt cells (code at 25 %: 0.162 vs 0.075). That is exactly the information hidden from an unprivileged daemon, because page-cache hits are invisible.

Pinning needs `CAP_IPC_LOCK`, so it was not used in the end-to-end runs. Data: `bench/results/phase7-olmoe/pinfrac-sweep/`.

6.3 Experiment B: what the real kernel does (and simulator validation)

The prose trace was replayed on the real model file through `mmap`, with 4 threads touching each expert's pages, under the memory wall. The trace needs 0.49 GB of expert data per token.

Budget	Variant	tokens/s	SSD read / token	Major faults / token
1.5 GB	kernel default	0.79	**1.80 GB**	956
1.5 GB	read-around off (`MADV_RANDOM`)	0.75	**0.16 GB**	38,802
1.5 GB	+ moepagerd (read-around on)	0.66	1.61 GB	881
1.5 GB	**read-around off + moepagerd**	**2.58**	0.05 GB	12,342
2.5 GB	kernel default	3.25	0.37 GB	338
2.5 GB	read-around off	**5.81**	0.02 GB	5,704
2.5 GB	+ moepagerd (read-around on)	2.82	0.36 GB	321
2.5 GB	read-around off + moepagerd	2.48	0.05 GB	12,641

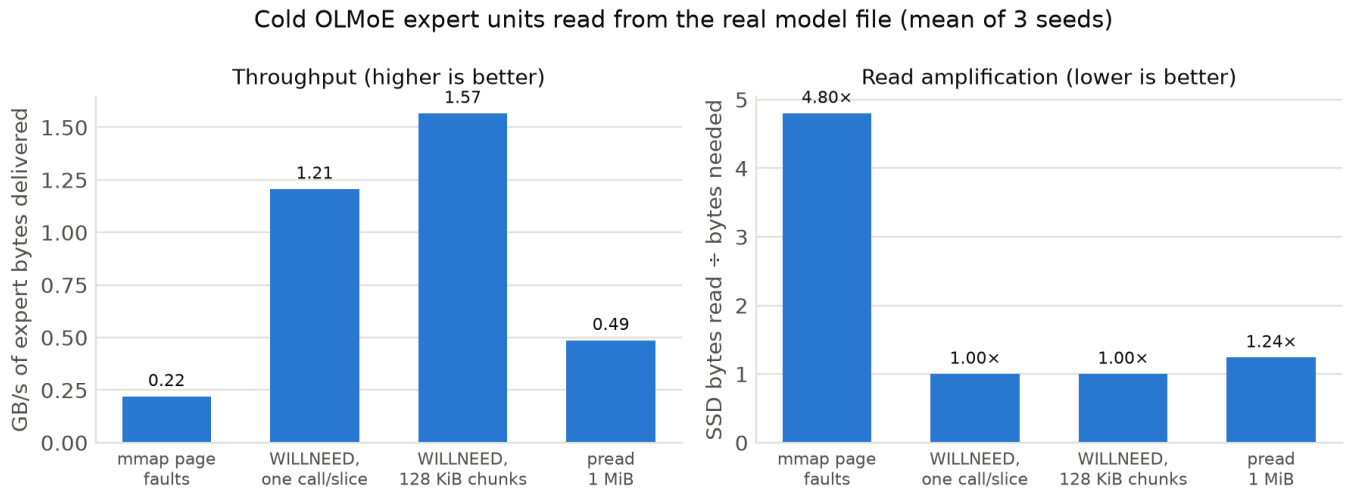


Reading: - **Read-around pollution dominates.** With the kernel's default 4 MiB read-around, the SSD read 1.80 GB per token at 1.5 GB to deliver 0.49 GB, and nearly every expert access missed: neighbouring, unused experts filled the cache. - **The simulator is validated once pollution is removed.** With read-around off, traffic fell 11× to 0.16 GB/token, which matches the simulator's LRU prediction for this capacity (0.14 GB/token, Section 6.2). - **Fewer bytes alone is not faster.** Read-around off produced 39,000 small 4 KiB faults per token, so speed stayed at 0.75 tok/s. - **moepagerd needs read-around off.** With read-around on, its sentinels fire on polluted pages and it made things slightly worse. With read-around off it turns the small faults into bulk reads: **3.3× faster** than the kernel default at 1.5 GB. - At 2.5 GB, where most accesses already hit, read-around off alone was best in this replay; the daemon added overhead.

6.4 Experiment C: fault-driven vs bulk read bandwidth

32 cold expert units (3.8 MB each, three slices) of the real model file were read in three ways, 3 seeds each:

Method	GB/s delivered	SSD bytes read ÷ needed
mmap page faults (8 threads)	0.15–0.24	4.8×
WILLNEED, one call per slice	0.97–1.42	1.00×
WILLNEED, 128 KiB chunks	**1.49–1.70**	**1.00×**
pread, 1 MiB blocks	0.44–0.56	1.24×

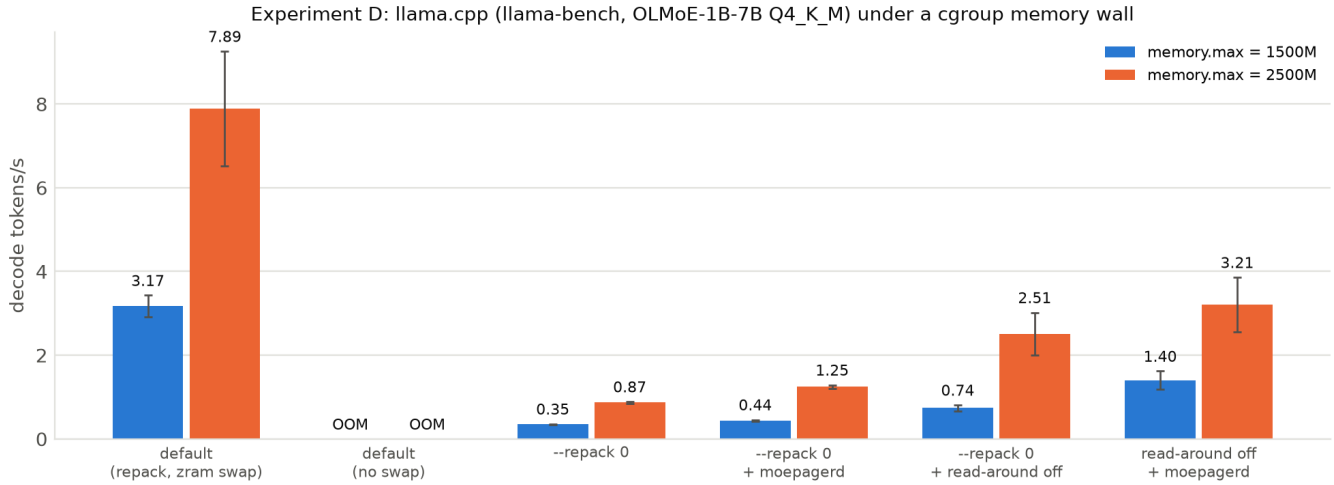


Bulk chunked readahead delivers expert bytes **≈ 7× faster** than page faults, with no amplification, so kill criterion **K2 does not fire**. Chunking matters: single calls are truncated by the kernel to the device readahead size.

6.5 Experiment D: end to end with llama.cpp

`llama-bench` decoded 32 tokens × 3 repetitions (after a 1-token warmup) inside the cgroup, starting cold. Decode throughput is mean ± standard deviation over the 3 repetitions. SSD traffic and major faults are cgroup/device totals divided by all 97 decoded tokens. PSI is the cgroup's memory-pressure "some" average over the last 10 s at exit.

Budget	Configuration	tokens/s	SSD GB / token	Major faults / token	Memory PSI (some, avg10)
1.5 GB	B0 default llama.cpp (repack; zram swap allowed)	3.17 ± 0.26	0.31	38,206	16.2 %
1.5 GB	B0S default llama.cpp, no swap	**OOM-killed**	—	—	—
1.5 GB	B2 `--repack 0` (page cache, kernel only)	0.35 ± 0.01	2.87	2,112	12.7 %
1.5 GB	D2 B2 + moepagerd	0.44 ± 0.02	2.90	871	4.8 %
1.5 GB	P B2 + read-around off (shim)	0.74 ± 0.07	0.25	50,418	18.1 %
1.5 GB	**PD B2 + read-around off + moepagerd**	**1.40 ± 0.22**	0.34	23,902	**1.0 %**
2.5 GB	B0 default llama.cpp (repack; zram swap allowed)	7.89 ± 1.37	0.13	13,854	12.0 %
2.5 GB	B0S default llama.cpp, no swap	**OOM-killed**	—	—	—
2.5 GB	B2 `--repack 0`	0.87 ± 0.02	1.19	1,402	14.8 %
2.5 GB	D2 B2 + moepagerd	1.25 ± 0.04	1.01	515	12.3 %
2.5 GB	P B2 + read-around off	2.51 ± 0.51	0.10	14,497	4.5 %
2.5 GB	**PD B2 + read-around off + moepagerd**	**3.21 ± 0.65**	0.17	8,161	2.5 %



Against the same engine serving weights from the page cache (B2): - PD is **4.0× faster at 1.5 GB** and **3.7× faster at 2.5 GB**. - PD reads **8.4× and 7.0× fewer bytes from the SSD** per token. - PD cuts memory-pressure stall (PSI) from 12.7 % to 1.0 % at 1.5 GB. This matters for other applications on the machine.

Splitting the gain: - Turning read-around off accounts for 2.1–2.9×. - moepagerd's completion readahead adds a further 1.9× (1.5 GB) and 1.3× (2.5 GB) on top. - moepagerd *without* read-around off (D2) helps only 1.3–1.4×.

Variability: PD at 2.5 GB was measured in three independent sessions: 3.30 (1 repetition), 5.42 ± 1.07 and 3.21 ± 0.65 tok/s. The table reports the last.

Against default llama.cpp (B0): - B0 is faster (3.17 and 7.89 tok/s), but it gets there by pushing **2.5 GB and 2.3 GB of repacked weights into zram**, compressed swap that lives in RAM and is not counted against `memory.max` (journal: "1.4G memory peak, 2.5G memory swap peak"). It therefore used far more than the budget. - With swap disabled, so it is held to the same budget (B0S), default llama.cpp was **OOM-killed at both budgets**: it cannot run at all. - moepager's configuration runs inside the budget.

A fair speed comparison against default llama.cpp therefore needs a machine whose swap is on disk, or a budget that includes zram. That is future work.

Ablation (refractory window): suppressing repeated completion requests for 20 ms looked like an obvious efficiency fix, but it doubled misses and cut PD at 2.5 GB from 3.21–5.42 to 1.09–2.05 tok/s. The repeats re-fetch pages reclaimed while the engine is still working through an expert. The window is therefore off by default.

6.6 Second model: gpt-oss-20b (experiments A–C)

The same pipeline was run on gpt-oss-20b MXFP4: 12.1 GB file, 24 layers × 32 experts, top-4, 13.25 MB per expert unit. Memory limits were 4 GB and 6 GB, and routing was recorded for 3 prompts × 128 tokens. Raw data: `bench/results/phase7-gptoss/`.

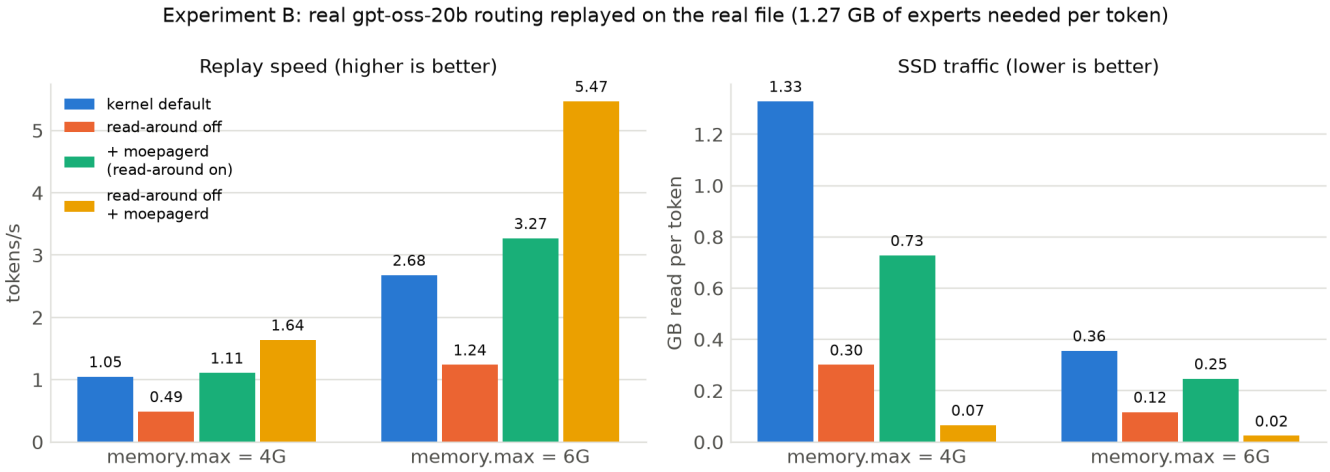
Experiment D did not run. Its first configuration, default llama.cpp, repacks 100 % of gpt-oss's expert bytes (≈ 10 GB) into anonymous memory and swaps them to zram, which lives in RAM. The machine ran critically low on memory and the run was stopped. This is itself evidence of the problem moepager avoids, but there are no end-to-end gpt-oss numbers.

- **Routing (A):** 40–47 % token-to-token reuse, next-layer recall 0.54–0.62 versus 0.39–0.46 for popularity. Belady* saves 44–67 % of LRU's misses at 15–50 % capacity, similar to OLMoE.
- **Bandwidth (C), 13.25 MB units:**

Method	GB/s	Amplification
mmap faults	0.14–0.44	≈ 2.5×
WILLNEED, 128 KiB chunks	1.44–1.88	1.00×
WILLNEED, one call per slice	0.93–1.10	1.44×, truncated
pread	0.31–0.65	—

- **Replay (B), prose trace, 1.27 GB of experts per token:**

Budget	Variant	tokens/s	SSD read / token
4 GB	kernel default	1.05	1.33 GB
4 GB	read-around off	0.49	0.30 GB
4 GB	+ moepagerd (read-around on)	1.11	0.73 GB
4 GB	**read-around off + moepagerd**	**1.64**	**0.07 GB**
6 GB	kernel default	2.68	0.36 GB
6 GB	read-around off	1.24	0.12 GB
6 GB	+ moepagerd (read-around on)	3.27	0.25 GB
6 GB	**read-around off + moepagerd**	**5.47**	**0.02 GB**



Reading: - The best configuration is the same as for OLMoE: read-around off plus completion readahead, **1.6× and 2.0× faster** than the kernel default, with **15–20× less SSD traffic**. - Two differences from OLMoE: - gpt-oss's expert slices (4.4 MB) are about as large as the read-around window, so read-around wastes less (≈ 1.05× at 4 GB). - Read-around off *alone* is slower than the default here, because 4 KiB faults replace useful large reads. - moepagerd turns those small faults back into bulk reads, and also helps somewhat on its own with read-around on (1.06× and 1.22×).

6.7 Comparison with expected outcomes and with existing methods

Expectation (IDEA_REVIEW, before measuring)	Outcome
Smarter eviction/pinning is the main lever (≥ 15 % fewer misses possible, K1)	The gap exists (Belady* saves 34–74 % vs LRU), but our online V(e) policy did not capture it reliably, and pinning needs privileges. **Not the main lever.**
Bulk readahead beats fault-driven reads (K2)	Confirmed: ≈ 7× (Exp. C).
Default llama.cpp is beaten (K3)	Only at equal memory, where it OOMs. With zram it is faster. **Partially.**
(not anticipated)	Fault read-around pollution is the dominant cost: 3.7–4.8× read amplification. Turning it off is required for both correctness of black-box detection and speed.

Relation to existing methods: - Si et al. found that `fadvise` lookahead hints gain ≤ 5 % on a GH200 with a `pread`-based engine. Our result differs because on an mmap-based engine the problem is *how* missing experts are read (small synchronous faults plus large wasted read-around), not *which* experts to keep. Completion readahead attacks exactly that. - In-engine systems (MoE-Infinity, HOBBIT, llama.cpp PR #25294) see router decisions directly and can avoid the issue entirely. moepager reaches a 3.7–4.0× improvement over the stock page-cache path without modifying the engine's code or binary, at the cost of one `LD_PRELOAD` hint.

6.8 Go / no-go decision

GO, with narrowed scope (recorded in PHASES.md): - The product is "read-around off + expert-completion readahead", validated end to end. - V(e) pinning and predictive prefetch are demoted to research options. - The next phase must validate on a second machine with disk swap and ext4, and on a larger model.

7. Conclusion

7.1 Summary

We built moepager, a toolchain and daemon that manages the Linux page cache for Mixture-of-Experts inference using the model file's structure, without changing the inference engine's code.

The project was run as an experiment with kill criteria fixed in advance. On a real model (OLMoE-1B-7B) on a 15 GB laptop, the outcome was:

- **Go on the mechanism, no-go on the original emphasis.** We expected smarter *eviction* to be the main lever. Measurement showed the dominant cost under memory pressure is the kernel's fault **read-around**. On this machine's btrfs it read 3.7× more data than needed and filled the cache with unused neighbouring experts.

- **What works:**

    - turning read-around off for the model mapping, with a 64-line `LD_PRELOAD` shim, since no stock llama.cpp option does this on single-node machines;
    - plus moepagerd's expert-completion readahead in 128 KiB chunks.

    Together they turn tens of thousands of 4 KiB faults per token into bulk reads. See Section 6.5 for the end-to-end numbers. - **What doesn't (yet):** - Completion readahead *without* turning read-around off made things worse. - Our V(e) pinning policy only matched LFU when allowed to pin most of the cache. With black-box (miss-only) observation it still loses 10–40 % in most cells, up to 2×, and it needs extra privileges. - Default llama.cpp is faster than every page-cache configuration *if* it may spill 2.3–2.5 GB of repacked weights into zram (compressed RAM). Without swap it is OOM-killed at both budgets. - **Smaller findings with independent value:** - The kernel silently truncates `WILLNEED`. - llama.cpp's default repacking moves 77–100 % of expert bytes out of the page cache. - llama.cpp's `--lazy-mode` keeps weights in anonymous memory. - btrfs reports different device numbers in `stat()` and `/proc/pid/maps`.

7.2 Limitations

- One machine, one filesystem (btrfs with a 4 MiB readahead window). End-to-end llama.cpp numbers exist only for OLMoE. gpt-oss-20b was evaluated with real routing, replay and the bandwidth benchmark, but its llama-bench run was stopped for low system memory. Qwen3-30B was only mapped from its header.
- Short runs (32 decode tokens × 3 repetitions in llama-bench); some configurations show large run-to-run spread.
- A desktop session was running during measurements.
- Pinning (`mlock`), demotion (`MADV_COLD`) and eBPF observation are implemented but untested, because they need privileges.
- The shim is still an intervention in the engine's process, although it doesn't change its code or binary.

7.3 Future work

1. Upstream: propose a llama.cpp option to apply `MADV_RANDOM` to expert tensors on single-node machines, which would remove the need for the shim.
2. Repeat on ext4 (128 KiB readahead) and on the larger models, with longer runs.

3. Evaluate pinning end to end with `CAP_IPC_LOCK` and a pin budget near 90 % of the expert cache. Narrow the miss-only gap with privileged hit observation (DAMON) where available.
4. Add a fallback when WILLNEED is ineffective (`MADV_POPULATE_READ`), and PSI-driven control of the memory budget to protect other applications.

8. References

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11. llama.cpp (ggml-org): source at commit 970b51c (`src/llama-mmap.cpp`, `src/llama-model-loader.cpp`, `ggml/src/ggml-cpu/repack.cpp`); PRs #25294, #26824, #29887, #29250; issue #20757; discussion #27149. https://github.com/ggml-org/llama.cpp
12. Linux kernel: `mm/readahead.c` (`force_page_cache_ra`), `include/trace/events/filemap.h`; man pages `mincore(2)`, `posix_fadvise(2)`, `madvise(2)`, `process_madvise(2)`, `cachestat(2)`.
13. GGUF model files: allenai/OLMoE-1B-7B-0924-GGUF, unsloth/Qwen3-30B-A3B-GGUF, ggml-org/gpt-oss-20b-GGUF (Hugging Face).